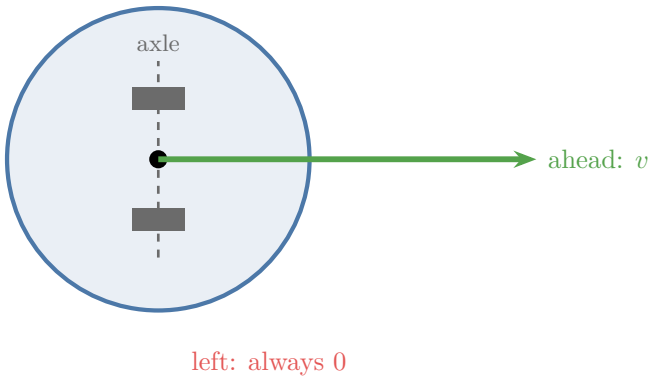
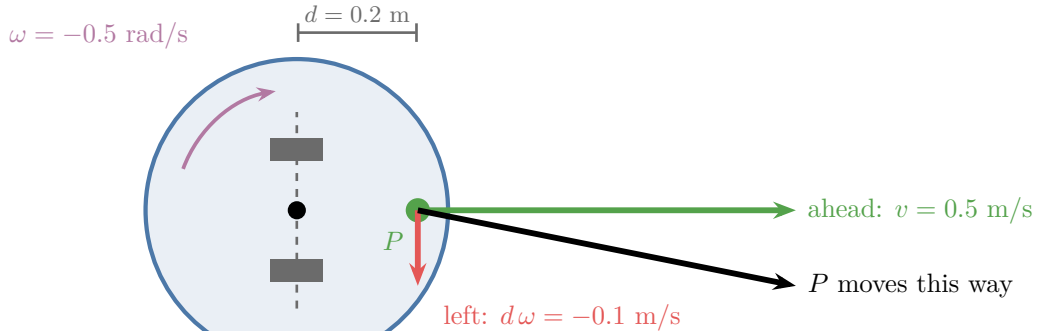
(a) the axle midpoint



it can only move the way the robot faces

(b) a point d ahead



turning swings P sideways, so P can move any way